Year 7

# REVISION PACK

## Unit 1

**Name:** _____

**Class:** _____

## Unit 1 · Averages and range

### Worked example: mode, median, mean and range

Pets owned by 8 students: 3 1 4 4 2 6 4 0

(a) Write down the mode. (b) Find the median. (c) Work out the mean. (d) Work out the range.

| | | | |
|---|---|---|---|
| (a) | Step 1 | Most common number | $4$ (3 times) |
| | Step 2 | Answer | $4$ |
| (b) | Step 1 | Put in order | $0, 1, 2, 3, 4, 4, 4, 6$ |
| | Step 2 | Middle two | $3$ and $4$ |
| | Step 3 | Answer | $(3 + 4) \div 2 = 3.5$ |
| (c) | Step 1 | Add all the numbers | $3 + 1 + 4 + 4 + 2 + 6 + 4 + 0 = 24$ |
| | Step 2 | Answer | $24 \div 8 = 3$ |
| (d) | Step 1 | Answer | $6 - 0 = 6$ |

## Practice: mode, median, mean and range

**1.1** Bags carried by 8 shoppers: 2 0 3 1 3 5 3 7

(a) Write down the mode. (b) Find the median. (c) Work out the mean. (d) Work out the range.

**1.2** Pets owned by 8 families: 1 4 2 4 0 4 3 6

(a) Write down the mode. (b) Find the median. (c) Work out the mean. (d) Work out the range.

**1.3** Goals scored in 8 football matches: 3 1 6 1 0 1 4 8

(a) Write down the mode. (b) Find the median. (c) Work out the mean. (d) Work out the range.

**1.4** Text messages sent by 8 students: 12 7 9 12 5 14 12 9

(a) Write down the mode. (b) Find the median. (c) Work out the mean. (d) Work out the range.

**1.5** Cars in 8 car parks: 20 25 18 25 30 22 25 19

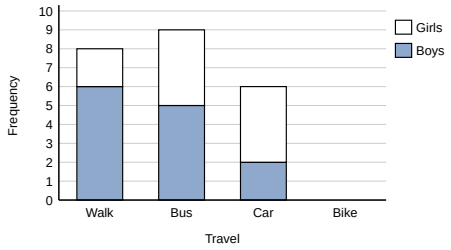
(a) Write down the mode. (b) Find the median. (c) Work out the mean. (d) Work out the range.

**1.6** Children in 8 families: 2 5 3 2 1 4 2 5

(a) Write down the mode. (b) Find the median. (c) Work out the mean. (d) Work out the range.

# Unit 1 · Compound bar charts

## Worked example: completing and reading a compound bar chart




Some students were asked how they travel to school. 3 boys and 4 girls come by bike.

(a) Complete the compound bar chart.

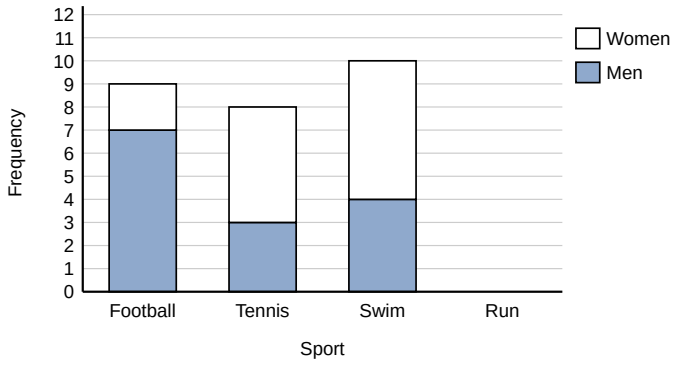
(b) How many girls come by bus?

(c) Work out the total number of boys.

| | | | |
|---|---|---|---|
| (a) | Step 1 | Draw the boys bar | up to 3 |
| | Step 2 | Girls on top | $3 + 4 = 7$ |
| | Step 3 | Answer | Bike bar: shade to 3, white from 3 to 7 |
| (b) | Step 1 | Top of the bar | 9 |
| | Step 2 | Top of the boys | 5 |
| | Step 3 | Answer | $9 - 5 = 4$ |
| (c) | Step 1 | Read each boys bar | $6,\ 5,\ 2,\ 3$ |
| | Step 2 | Answer | $6 + 5 + 2 + 3 = 16$ |

## Practice: completing and reading a compound bar chart

**2.1**



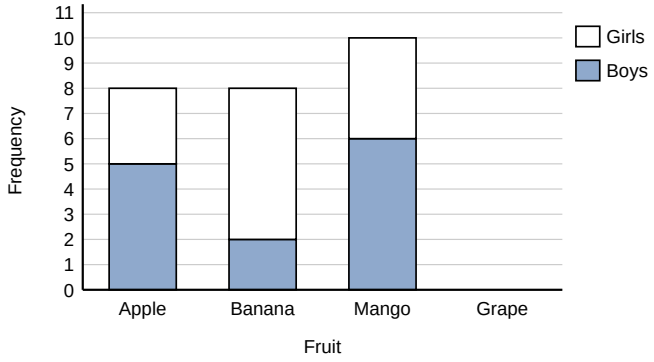

People were asked what sport they play. 5 men and 3 women chose run.

(a) Complete the compound bar chart.

(b) How many women chose swim?

(c) Work out the total number of men.

**2.2**




Students were asked their favourite fruit. 3 boys and 4 girls chose grape.

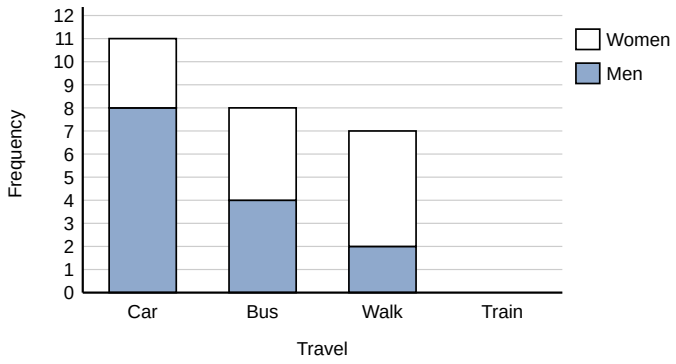
(a) Complete the compound bar chart.

(b) How many girls chose banana?

(c) Work out the total number of boys.

## Practice: completing and reading a compound bar chart (continued)

**2.3**



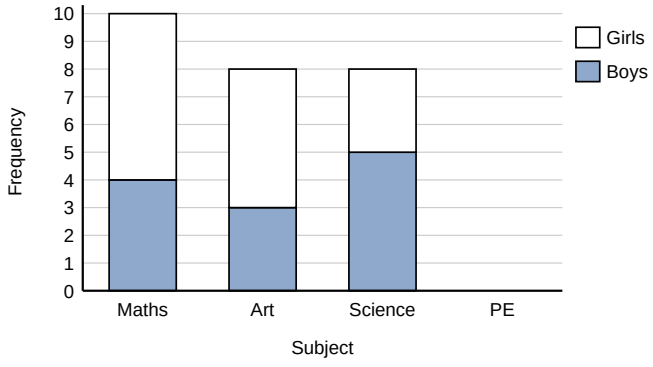

People were asked how they get to work. 3 men and 6 women chose train.

(a) Complete the compound bar chart.

(b) How many women chose car?

(c) Work out the total number of men.

**2.4**




Students were asked their favourite subject. 6 boys and 2 girls chose pe.

(a) Complete the compound bar chart.

(b) How many girls chose art?

(c) Work out the total number of boys.

### Worked example: the mean and adding a new value

Sweets in 5 bags: 8 5 7 4 6

(a) Work out the mean. (b) A sixth bag has 2 sweets. Will the mean for all 6 bags be greater or less? Give a reason.

| | | | |
|---|---|---|---|
| (a) | Step 1 | Add | $8 + 5 + 7 + 4 + 6 = 30$ |
| | Step 2 | Divide by 5 | $30 \div 5 = 6$ |
| | Step 3 | Answer | $6$ |
| (b) | Step 1 | Compare 2 with the mean | $2 < 6$ |
| | Step 2 | Answer | Less, because 2 is less than the mean |

## Practice: the mean and adding a new value

**3.1** Children in 5 houses: 3 1 5 2 4

(a) Work out the mean. (b) A sixth house has 0. Will the mean for all 6 be greater or less? Give a reason.

**3.2** Goals in 5 matches: 2 4 1 3 5

(a) Work out the mean. (b) A sixth match has 6. Will the mean for all 6 be greater or less? Give a reason.

**3.3** Books read by 5 students: 6 9 4 7 9

(a) Work out the mean. (b) A sixth student has 1. Will the mean for all 6 be greater or less? Give a reason.

**3.4** Marks of 5 students: 12 15 10 14 9

(a) Work out the mean. (b) A sixth student has 20. Will the mean for all 6 be greater or less? Give a reason.

**3.5** Eggs from 5 hens: 5 7 4 6 8

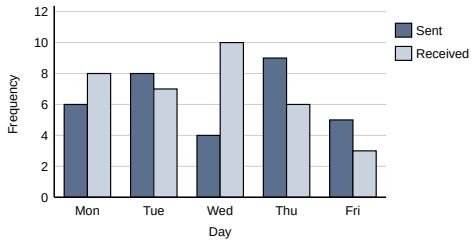
(a) Work out the mean. (b) A sixth hen has 2. Will the mean for all 6 be greater or less? Give a reason.

**3.6** Minutes late for 5 trains: 4 0 6 3 2

(a) Work out the mean. (b) A sixth train has 9. Will the mean for all 6 be greater or less? Give a reason.

# Unit 1 · Dual bar charts

## Worked example: reading and comparing a dual bar chart




The dual bar chart shows the emails Omar sent and received on 5 days.

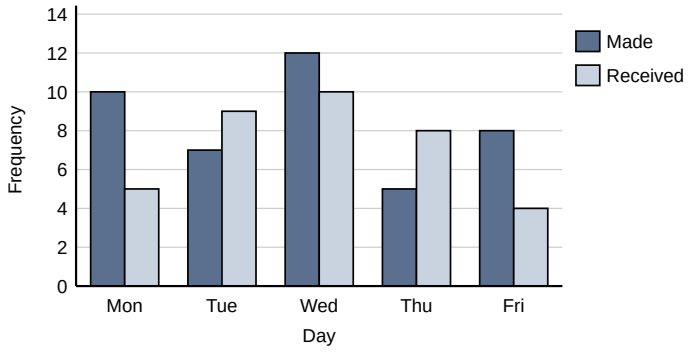
(a) On which day was the difference between sent and received the greatest?

(b) Omar says, "I sent more emails than I received." Is Omar correct? Show your working.

| | | | |
|---|---|---|---|
| (a) | Step 1 | Find each difference | $2,\ 1,\ 6,\ 3,\ 2$ |
| | Step 2 | Answer | Wednesday |
| (b) | Step 1 | Add the sent | $6 + 8 + 4 + 9 + 5 = 32$ |
| | Step 2 | Add the received | $8 + 7 + 10 + 6 + 3 = 34$ |
| | Step 3 | Compare | $32 < 34$ |
| | Step 4 | Answer | No, he received more |

## Practice: reading and comparing a dual bar chart

**4.1**

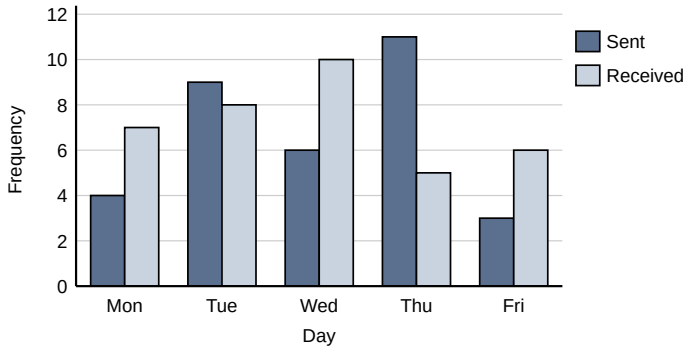



The dual bar chart shows the number of calls Jamila made and received on 5 days.

(a) On which day was the difference between made and received the greatest?

(b) Jamila says, "I made more calls than I received." Is Jamila correct? Show your working.

**4.2**



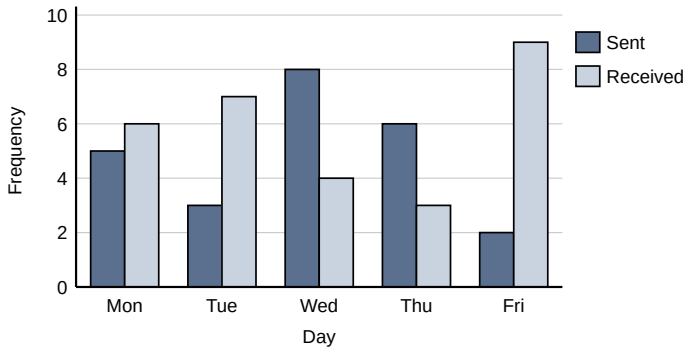

The dual bar chart shows the number of texts Sami sent and received on 5 days.

(a) On which day was the difference between sent and received the greatest?

(b) Sami says, "I sent more texts than I received." Is Sami correct? Show your working.

## Practice: reading and comparing a dual bar chart (continued)

**4.3**

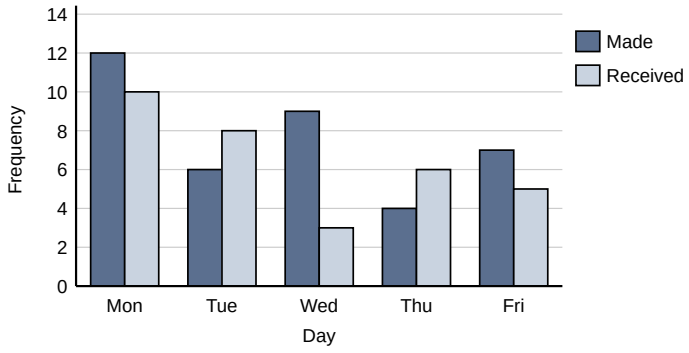



The dual bar chart shows the number of emails Lina sent and received on 5 days.

(a) On which day was the difference between sent and received the greatest?

(b) Lina says, "I sent fewer emails than I received." Is Lina correct? Show your working.

**4.4**




The dual bar chart shows the number of calls Yusuf made and received on 5 days.

(a) On which day was the difference between made and received the greatest?

(b) Yusuf says, "I made more calls than I received." Is Yusuf correct? Show your working.

### Worked example: comparing with the median and range

Sam's points in 7 games: 4 8 5 10 6 7 5

(a) Work out the median and the range.

(b) Ella: median 8, range 3. Compare Sam and Ella.

| | | | |
|---|---|---|---|
| (a) | Step 1 | Put in order | $4, 5, 5, 6, 7, 8, 10$ |
| | Step 2 | Median: the middle | $6$ |
| | Step 3 | Range: biggest − smallest | $10 - 4 = 6$ |
| (b) | Step 1 | Compare the medians | $8 > 6$ |
| | Step 2 | Say what it means | Ella did better on average |
| | Step 3 | Compare the ranges | $3 < 6$ |
| | Step 4 | Answer | Ella is more consistent |

## Practice: comparing with the median and range

**5.1** Kareem's points in 7 basketball games: 5 2 9 6 4 8 6

(a) Work out the median and the range.

(b) Aziz: median 8, range 15. Compare Kareem and Aziz.

**5.2** Hana's points in 7 netball games: 14 18 11 16 20 13 17

(a) Work out the median and the range.

(b) Rania: median 12, range 4. Compare Hana and Rania.

**5.3** Tariq's goals in 7 football matches: 2 6 3 8 5 1 4

(a) Work out the median and the range.

(b) Faris: median 6, range 3. Compare Tariq and Faris.

**5.4** Dana's marks in 7 quizzes: 30 25 34 28 40 27 31

(a) Work out the median and the range.

(b) Mona: median 26, range 20. Compare Dana and Mona.

**5.5** Ibrahim's points in 7 basketball games: 9 13 10 7 12 15 11

(a) Work out the median and the range.

(b) Khalid: median 9, range 3. Compare Ibrahim and Khalid.

**5.6** Nadia's goals in 7 hockey matches: 6 8 4 9 7 5 10

(a) Work out the median and the range.

(b) Salma: median 9, range 12. Compare Nadia and Salma.

## Unit 1 · Grouped data

**Worked example: grouped frequency tables**

The heights of 12 plants, in cm: 23 35 18 27 31 14 26 38 22 29 25 33

(a) Complete the frequency table. Use groups of width 10.

(b) Draw a frequency diagram.

(c) Write down the modal class.

(d) How many are more than 20 cm?

| | | | |
|---|---|---|---|
| (a) | **Step 1** | Tally each number | $10 < h \leq 20:\ 2$ |
| | **Step 2** | | $20 < h \leq 30:\ 6$ |
| | **Step 3** | | $30 < h \leq 40:\ 4$ |
| (b) | **Step 1** | Bars, no gaps | heights $2,\ 6,\ 4$ |
| (c) | **Step 1** | Biggest frequency | $20 < h \leq 30$ |
| (d) | **Step 1** | Add the groups above 20 | $6 + 4 = 10$ |
| | **Step 2** | Answer | $10$ |

### Practice: grouped frequency tables

**6.1** The masses of 15 apples, in g:

112 145 130 168 155 121 139 147 176 163 158 141 126 149 137

| M (g) | Tally | Frequency |
|---|---|---|
| 100 < M ≤ 120 | | |
| | | |
| | | |
| | | |

(a) Complete the frequency table. Use groups of width 20.

(b) Draw a frequency diagram.

(c) Write down the modal class.

(d) How many are more than 140 g?

**6.2** The times taken by 16 runners, in minutes:

23 35 41 28 37 32 45 26 39 31 34 22 38 43 36 29

| T (minutes) | Tally | Frequency |
|---|---|---|
| 20 < T ≤ 30 | | |
| | | |
| | | |

(a) Complete the frequency table. Use groups of width 10.

(b) Draw a frequency diagram.

(c) Write down the modal class.

(d) How many are more than 30 minutes?

## Practice: grouped frequency tables (continued)

**6.3** The masses of 18 potatoes, in g:

212 358 305 245 419 337 315 320 284 180 312 390 187 289 336 340 332 310

| M (g) | Tally | Frequency |
|---|---|---|
| 100 < M ≤ 200 | | |
| | | |
| | | |
| | | |

(a) Complete the frequency table. Use groups of width 100.

(b) Draw a frequency diagram.

(c) Write down the modal class.

(d) How many are more than 300 g?

**6.4** The heights of 15 students, in cm:

142 156 149 161 153 138 157 165 151 147 159 154 144 152 158

| H (cm) | Tally | Frequency |
|---|---|---|
| 130 < H ≤ 140 | | |
| | | |
| | | |
| | | |

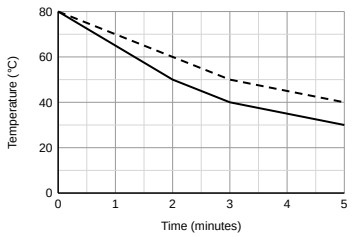
(a) Complete the frequency table. Use groups of width 10.

(b) Draw a frequency diagram.

(c) Write down the modal class.

(d) How many are more than 150 cm?

### Worked example: reading and comparing line graphs




Two cups of water cool down. The graph shows their temperature every minute. — Cup A - - - Cup B

(a) Write down the temperature at the start.

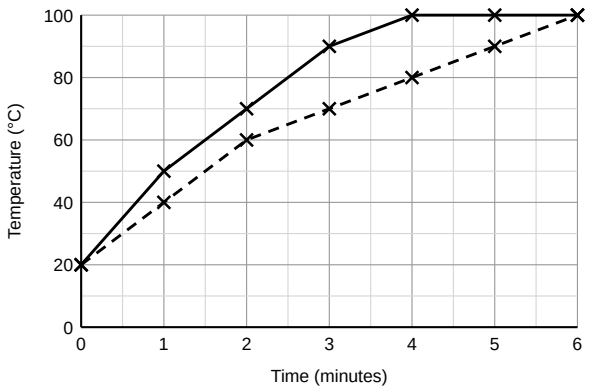
(b) Compare the times taken for each cup to reach 40 °C.

(c) Work out the difference in temperature after 2 minutes.

| | | | |
|---|---|---|---|
| (a) | Step 1 | Read at 0 minutes | $80\ ^\circ\text{C}$ |
| (b) | Step 1 | Cup A reaches 40 | $3$ minutes |
| | Step 2 | Cup B reaches 40 | $5$ minutes |
| | Step 3 | Answer | A is 2 minutes quicker |
| (c) | Step 1 | Read both at 2 minutes | $50$ and $60$ |
| | Step 2 | Answer | $60 - 50 = 10\ ^\circ\text{C}$ |

## Practice: reading and comparing line graphs

**7.1**




Two liquids are heated from room temperature. The line graph shows their temperature every minute.

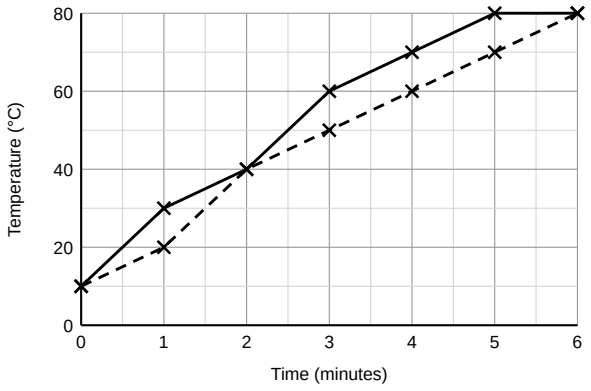
— Liquid A - - - Liquid B

(a) Write down the temperature at the start.

(b) Compare the times taken for each liquid to reach 100 °C.

(c) Work out the difference in temperature after 3 minutes.

**7.2**




Two liquids are heated. The line graph shows their temperature every minute.

— Liquid A - - - Liquid B

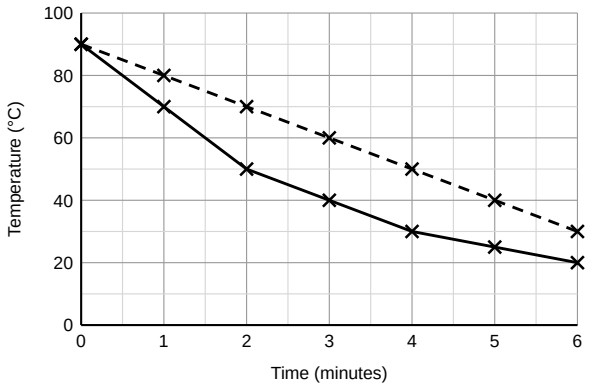
(a) Write down the temperature at the start.

(b) Compare the times taken for each liquid to reach 60 °C.

(c) Work out the difference in temperature after 2 minutes.

## Practice: reading and comparing line graphs (continued)

**7.3**




Two liquids are left to cool. The line graph shows their temperature every minute.

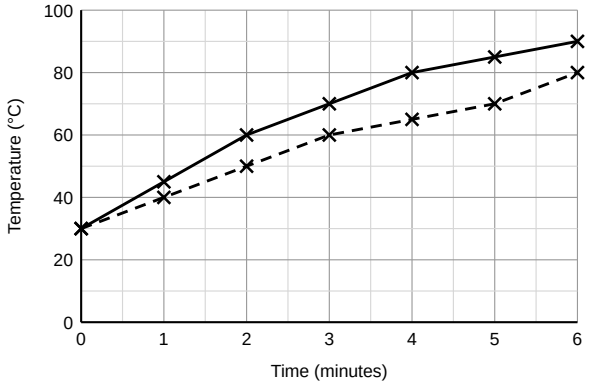
— Liquid A  - - - Liquid B

(a) Write down the temperature at the start.

(b) Compare the times taken for each liquid to reach 50 °C.

(c) Work out the difference in temperature after 4 minutes.

**7.4**




Two liquids are heated. The line graph shows their temperature every minute.

— Liquid A  - - - Liquid B

(a) Write down the temperature at the start.

(b) Compare the times taken for each liquid to reach 60 °C.

(c) Work out the difference in temperature after 4 minutes.